

**Victorian Certificate of Education
2016****PHYSICS****Written examination****Wednesday 9 November 2016****Reading time: 9.00 am to 9.15 am (15 minutes)****Writing time: 9.15 am to 11.45 am (2 hours 30 minutes)****FORMULA SHEET****Instructions**

- A question and answer book is provided with this formula sheet.

Students are NOT permitted to bring mobile phones and/or any other unauthorised electronic devices into the examination room.

1	velocity; acceleration	$v = \frac{\Delta x}{\Delta t}; \quad a = \frac{\Delta v}{\Delta t}$
2	equations for constant acceleration	$v = u + at$ $x = ut + \frac{1}{2}at^2$ $v^2 = u^2 + 2ax$ $x = \frac{1}{2}(v + u)t$
3	Newton's second law	$\Sigma F = ma$
4	circular motion	$a = \frac{v^2}{r} = \frac{4\pi^2 r}{T^2}$
5	Hooke's law	$F = -kx$
6	elastic potential energy	$\frac{1}{2}kx^2$
7	gravitational potential energy near the surface of Earth	mgh
8	kinetic energy	$\frac{1}{2}mv^2$
9	Newton's law of universal gravitation	$F = G \frac{M_1 M_2}{r^2}$
10	gravitational field	$g = G \frac{M}{r^2}$
11	acceleration due to gravity at Earth's surface	$g = 10 \text{ m s}^{-2}$
12	voltage; power	$V = RI \quad P = VI = I^2 R$
13	resistors in series	$R_T = R_1 + R_2$
14	resistors in parallel	$\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2}$
15	transformer action	$\frac{V_1}{V_2} = \frac{N_1}{N_2}$
16	AC voltage and current	$V_{\text{RMS}} = \frac{1}{\sqrt{2}} V_{\text{peak}} \quad I_{\text{RMS}} = \frac{1}{\sqrt{2}} I_{\text{peak}}$
17	magnetic force	$F = I l B$

18	electromagnetic induction	emf: $\varepsilon = -N \frac{\Delta \Phi}{\Delta t}$ flux: $\Phi = BA$
19	transmission losses	$V_{\text{drop}} = I_{\text{line}} R_{\text{line}}$ $P_{\text{loss}} = I_{\text{line}}^2 R_{\text{line}}$
20	mass of the electron	$m_e = 9.1 \times 10^{-31} \text{ kg}$
21	charge on the electron	$e = -1.6 \times 10^{-19} \text{ C}$
22	Planck's constant	$h = 6.63 \times 10^{-34} \text{ J s}$ $h = 4.14 \times 10^{-15} \text{ eV s}$
23	speed of light	$c = 3.0 \times 10^8 \text{ m s}^{-1}$
24	photoelectric effect	$E_{K \text{ max}} = hf - W$
25	photon energy	$E = hf$
26	photon momentum	$p = \frac{h}{\lambda}$
27	de Broglie wavelength	$\lambda = \frac{h}{p}$
28	speed, frequency and wavelength	$v = f\lambda$
29	energy transformations for electrons in an electron gun (<100 keV)	$\frac{1}{2}mv^2 = eV$
30	radius of electron path	$r = \frac{mv}{eB}$
31	magnetic force on a moving electron	$F = evB$
32	Bragg's law	$n\lambda = 2d \sin \theta$
33	electric field between charged plates	$E = \frac{V}{d}$
34	band gap energy	$E = \frac{hc}{\lambda}$
35	Snell's law	$n_1 \sin \theta_1 = n_2 \sin \theta_2$
36	intensity and level	sound intensity level (in dB) $L(\text{dB}) = 10 \log_{10} \left(\frac{I}{I_0} \right)$ where $I_0 = 1.0 \times 10^{-12} \text{ W m}^{-2}$

37	Lorentz factor	$\gamma = \frac{1}{\sqrt{1 - \frac{v^2}{c^2}}}$
38	time dilation	$t = t_0 \gamma$
39	length contraction	$L = \frac{L_0}{\gamma}$
40	relativistic mass	$m = m_0 \gamma$
41	total energy	$E_{\text{total}} = E_k + E_{\text{rest}} = mc^2$
42	stress	$\sigma = \frac{F}{A}$
43	strain	$\varepsilon = \frac{\Delta L}{L}$
44	Young's modulus	$E = \frac{\text{stress}}{\text{strain}}$
45	capacitors	time constant : $\tau = RC$
46	universal gravitational constant	$G = 6.67 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$
47	mass of Earth	$M_E = 5.98 \times 10^{24} \text{ kg}$
48	radius of Earth	$R_E = 6.37 \times 10^6 \text{ m}$
49	mass of the electron	$m_e = 9.1 \times 10^{-31} \text{ kg}$
50	charge on the electron	$e = -1.6 \times 10^{-19} \text{ C}$
51	speed of light	$c = 3.0 \times 10^8 \text{ m s}^{-1}$

Prefixes/Unitsp = pico = 10^{-12} n = nano = 10^{-9} μ = micro = 10^{-6} m = milli = 10^{-3} k = kilo = 10^3 M = mega = 10^6 G = giga = 10^9 t = tonne = 10^3 kg **END OF FORMULA SHEET**